

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20							***** MODULAR VALVES *****			B1	ALL BOLTS ARE METRIC DIA.
30										B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
30			MODULAR DBB	K01AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0		
40		VAV	MODULAR DBB	K03AM10	3	3	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - FULL SIZE TYPE 2BV	0	0		
50			MODULAR DBB	M01AM10	0.5	2	MODULAR VALVE DB&B BALL TYPE 150# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
60	U04	FS1	MODULAR DBB	M03AM10	0.75	2	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	FS1	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
70		VAW	MODULAR DBB	M03EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2	GA6	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
80		VAW	MODULAR DBB	M03EM10	1	1	MODULAR VALVE DB&B BALL TYPE 300# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	ND8	USE ONLY WHERE SPACE IS LIMITED AND WITH APPROVAL BY MATERIALS ENGINEER.
90		VB2					***** BALL VALVES *****			UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
90											
100											
100			BALL	A01AM09	3	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0		
110	U04		BALL	A01AR09	4	4	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0		
120	U04		BALL	A01AR15	6	16	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT RB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
130			BALL	B01AM09	0.75	2	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0		
140		VA5	BALL	B01AM09	3	3	BALL 150# RF INC 825 INC TRIM 625 FLOATING SOFT FB LO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
150	U04	VA5	BALL	B01AR11	4	4	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 FLOATING SOFT FB GO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
160	U04	VA5	BALL	B01AR15	6	16	BALL 150# RF LTCS + CLAD INC 825 INC TRIM 625 TRUNNION SOFT FB GO	0	0		
170		FS1	BALL	B03AM09	0.75	2	BALL 300# RF INC 825 INC 625 TRIM FLOATING SOFT FB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
180		FS1	BALL	A03AM09	3	3	BALL 300# RF INC 825 INC TRIM 625 FLOATING SOFT RB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
190							***** BUTTERFLY VALVES *****				
190											
200			BUTTERFLY	T01AR63	10	20	BUTTERFLY 150# RF LTCS + CLAD INC 825 INC TRIM 625 TOV METAL DBL FLG SHORT PAT GO	0	0		
210							***** GLOBE VALVES *****				
210											
220											
220			GLOBE	L01AM49	0.75	3	GLOBE 150# RF INC 825 INC TRIM 625 OS&Y HW	0	0		
230			GLOBE	L01AR49	4	6	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y HW	0	0		
240			GLOBE	L01AR44	8	10	GLOBE 150# RF LTCS + CLAD INC 825 INC TRIM 625 OS&Y GO	0	0		
250							***** CHECK VALVES *****				
250											
260											
260		VC1	CHECK	C01AM51	0.75	1.5	CHECK PISTON 150# RF INC 825 INC TRIM 625	0	0		
270	U04		CHECK	C01AM46	2	3	CHECK DUAL WAFER LUG 150# RF INC 825 TRIM INC 625 THROU BOLTED	0	0		
280	U04		CHECK	C01AR46	4	6	CHECK DUAL WAFER LUG 150# RF LTCS + CLAD INC 825 TRIM INC 625 THROU BOLTED	0	0		
290	U04		CHECK	C01AR53	8	16	CHECK DUAL DBL FLG 150# RF LTCS + CLAD INC 825 INC TRIM 625	0	0		
300											
300	U04		GATE	G01AM49	1	1	GATE 150# RF INC 825 INC TRIM 625 WEDGE HW	0	0		
310							***** PIPE *****				
310											
320											
320		812	PIPE	5358511	0.75	1.5	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0		
330			PIPE	5358513	2	2	PIPE SCH XXS SMLS LTCS A333 GR.6 BE	0	0		
340			PIPE	5358514	3	16	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0		
350			PIPE	5358577	20	20	PIPE SCH XS EFW LTCS A671 CC60 CL.22 BE	0	0		
360							***** FITTINGS *****				
360											
370											
370			ELBOW	5392870	0.75	2	ELL 90 DEG LR XXS LTCS A420 WPL6	0	0		
380			ELBOW	5392600	3	16	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6	0	0		
390	U04		ELBOW SR	5392247	6	8	ELL BW 90 DEG SR SCH XS LTCS A420 WPL6	0	0		
400	U04		ELBOW SR	5392247	12	12	ELL BW 90 DEG SR SCH XS LTCS A420 WPL6	0	0		
410			ELBOW	5392876	0.75	2	ELL 45 DEG LR XXS LTCS A420 WPL6	0	0		
420			ELBOW	5392606	3	20	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6	0	0		
430											
430			TEE	5392879	0.75	2	TEE SCH XXS LTCS A420 WPL6	0	0		
440			TEE	5392609	3	16	TEE BW SCH XS LTCS A420 WPL6	0	0		
450			TEE RED	5392877	0.75	1.5	TEE RED SCH XXS LTCS A420 WPL6	1	2		
460			TEE RED	5392897	1.5	2	TEE RED BW SCH XS/XXS LTCS A420 WPL6	3	4		
470			TEE RED	5392661	3	14	TEE RED BW SCH XS LTCS A420 WPL6	4	16		
480			TEE RED	5392413	16	16	TEE RED SCH XS LTCS A420 WPL6-W	20	20		
490											
490	U04		CAP	5392880	0.75	2	CAP BW SCH XXS LTCS A420 WPL6	0	0		
500			CAP	5392610	3	16	CAP BW SCH XS LTCS A420 WPL6	0	0		
510											
510			RED CONC	5392881	0.75	1.5	REDUCER CONC SCH XXS LTCS A420 WPL6	1	2		
520			RED CONC	5392898	1.5	2	REDUCER CONC SCH XS/XXS LTCS A420 WPL6	3	4		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 6 MM	SOUR SERVICE - NACE MR0175 / ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 150 RF TEMPERATURE RANGE: -40 / 200 °C	12/16/2024	BY OZAC	U04	Page 1 of 2	150K23

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530			RED CONC	5392458	3	14	REDUCER CONC SCH XS LTCS A420 WPL6	4	16	
540			RED ECC	5392882	0.75	1.5	REDUCER ECC SCH XXS LTCS A420 WPL6	1	2	
550			RED ECC	5392899	1.5	2	REDUCER ECC SCH XS/XXS LTCS A420 WPL6	3	4	
560			RED ECC	5392461	3	14	REDUCER ECC SCH XS LTCS A420 WPL6	4	16	
570										
570	U04		OLETS	5504211	0.75	1.5	NIPOFLANGE 150# RF SCH XXS LTCS A350 LF2 CL.1	2	2	
580	U04		OLETS	5504212	0.75	2	NIPOFLANGE 150# RF SCH XS/XXS LTCS A350 LF2 CL.1	3	16	
590	U04		OLETS	5504215	1	1.5	NIPOFLANGE 300# RF SCH XXS LTCS A350 LF2 CL.1	1.5	2	
600	U04		OLETS	5504216	0.75	2	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	3	16	
610	U04	UJ2	OLETS	5504213	1.5	1.5	NIPOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	4	16	
620										
620	U04		OLETS	5505197	0.75	1	WELDOLET SCH XXS LTCS A350 LF2 CL.1	2	2	
630	U04		OLETS	5505180	0.75	2	WELDOLET SCH XS/XXS LTCS A350 LF2 CL.1	3	16	
640	U04		OLETS	5505335	3	4	WELDOLET SCH XS LTCS A350 LF2 CL.1	8	16	
650										
650	U04	UJ2	ELBOFLG	5516762	1.5	1.5	ELBOFLANGE 300# RF SCH XS/XXS LTCS A350 LF2 CL.1	4	16	
660										
660	U04	ND8		5509807	1.5	1.5	ELBOLET WT XS/XXS LTCS A350 LF2 CL.1	12	12	
670										
670		FS1					***** FLANGES *****			
680										
680	U04	FA8	FLANGE	5599692	0.5	0.5	FLG SCRDR (FNPT) REDUCING 150# RF LTCS A350 LF2 CL.1	0.75	2	
690	U04	FA8	TEE RED	5599703	0.5	0.5	FLG SCRDR (FNPT) REDUCING 300# RF LTCS A350 LF2 CL.1	0.75	0.75	
700										
700			FLANGE	5851200	0.5	2	FLANGE ADAPTER 150# RF 316 SS X 12 MM TUBE	0	0	
710			FLANGE	5851203	0.75	0.75	FLANGE ADAPTER 300# RF 316 SS X 12 MM TUBE	0	0	
720										
720	U04		FLANGE	5556221	0.75	2	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
730	U04		FLANGE	5556264	3	20	FLG WN 150# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
740	U04		FLANGE	5556481	0.75	2	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
750	U04		FLANGE	5556476	3	16	FLG WN 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
760	U04		FLANGE	5556976	0.75	2	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
770	U04		FLANGE	5557036	3	16	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
780	U04		FLANGE	5557141	6	8	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
790	U04		BLIND	5615566	0.75	16	FLG BLIND 150# RF LTCS A350 LF2 CL.1	0	0	
800	U04		BLIND	5615576	0.75	16	FLG BLIND 300# RF LTCS A350 LF2 CL.1	0	0	
810										
810							***** ORIFICE FLANGES *****			
820										
820	U04		ORIFICE FLG	5623950	2	2	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XXS	0	0	
830	U04		ORIFICE FLG	5623947	3	16	WN ORIF SET 300# RF LTCS A350 LF2 CL.1 SCH XS	0	0	
840										
840							***** PIPE NIPPLES *****			
850										
850		812		0			NIPPLE PBE - CUT FROM PIPE			
860										
860		FS1					***** GASKETS *****			
870										
870			GASKET	5673043	0.75	20	GASKET 150# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
880			GASKET	5673033	0.75	16	GASKET 300# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
890	U04	GA6		5673017	2	2	GASKET 300# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
900			GASKET	5673103	0.75	16	GASKET 600# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
910			GASKET	5673117	6	8	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
920										
920		B1					***** BOLTING *****			
930										
930	U04		BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
940										
940							***** BLINDS & SPACERS *****			
950										
950			SPEC BLIND	5749580	0.75	14	SPECTACLE BLIND 150# RF LTCS A 516 GR.70	0	0	
960			SPEC BLIND	5749581	1.5	10	SPECTACLE BLIND 300# RF LTCS A516 GR.70	0	0	
970										
970			PADDLE	5750023	16	20	PADDLE BLANK 150# RF LTCS A516 GR.70	0	0	
980			PADDLE	5750100	12	16	PADDLE BLANK 300# RF LTCS A516 GR.70	0	0	
990			PADDLE	5750240	16	20	PADDLE SPACER 150# RF LTCS A516 GR.70	0	0	
1000			PADDLE	5750200	12	16	PADDLE SPACER 300# RF LTCS A516 GR.70	0	0	
1010										

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